

Final Year Project Proposal

**ContracthubPk: A Secure Web Platform for Managing Freelance Contracts
and Escrow Payments using Local Payment Gateways**

**Submitted to the
Department of Software Engineering
University Of Rasul, Mandi Bahaud Din**

**in Partial Fulfillment of
the Requirements for the Degree of
Bs Software Engineering
By**

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Date: _____

Project Proposal

Class: BS Software Engineering

Title: ContracthubPk: A Secure Web Platform for Managing Freelance Contracts and Escrow Payments using Local Payment Gateways

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Problem Statement: In Pakistan’s freelance market, most freelancers and clients rely on informal communication channels such as social media or messaging apps to negotiate and manage projects. This often leads to trust issues, payment fraud, lack of formal contracts, and project disputes. Existing international platforms like Fiverr and Upwork charge high commissions and do not support local payment gateways, making them less accessible for many users. Therefore, there is a need for a secure, automated web platform that allows freelancers and clients to create digital contracts, manage escrow based payments, and ensure transparent collaboration using local payment methods such as Easypaisa and JazzCash.

Introduction: The proposed project, titled “A Secure Web Platform for Managing Freelance Contracts and Escrow Payments using Local Payment Gateways,” aims to address the growing issues of trust, payment security, and contract management within Pakistan’s local freelance market. Many freelancers and clients in Pakistan prefer to work outside international platforms like Fiverr and Upwork due to their high service charges and intense competition. However, working independently through social media or informal communication channels often leads to payment disputes, project delays, and a lack of accountability, as there is no standardized system to ensure safe transactions or enforce digital agreements.

This project proposes a web-based platform that enables freelancers and clients to create, manage, and execute freelance contracts securely while ensuring fair and transparent payments through escrow mechanisms. The system will integrate local payment gateways such as Easypaisa, JazzCash, and Visa/MasterCard to support seamless and accessible financial transactions. Key features will include contract creation, milestone-based payments, automated escrow release, dispute resolution, and user verification to ensure reliability and confidence between both parties.

By automating the freelance workflow and ensuring secure payment management, this system will serve as a trusted digital bridge between freelancers and clients in Pakistan. It will promote digital trust, financial inclusion, and transparent collaboration, while reducing the dependency on costly international freelancing platforms.

Purpose: to develop a secure web platform that enables freelancers and clients to manage contracts and payments through an automated escrow system. It aims to eliminate trust issues and payment fraud by integrating local gateways such as Easypaisa and JazzCash.

Scope: The scope of ContracthubPk is limited to designing and implementing a secure web based freelance contract and escrow management system for Pakistan's local market within an academic environment.

Objectives:

- Connecting freelancers and clients for transparent collaboration.
- Implemented an automated escrow system to ensure safe and milestone-based payments
- Integrate local payment gateways like Easypaisa, JazzCash
- Enable digital contract creation and management for clarity
- Provide a dispute resolution and review system

Functional Requirements:

1. The system allows users to create an account and log in securely as a **client** or **freelancer**.
2. Users can create and update their personal profiles, including basic information and payment details.
3. Both clients and freelancers can create a **digital contract** by defining scope, budget, and milestones.
4. The system enables users to view, accept, or reject contract requests.
5. Clients can deposit contract payments using **local payment methods** such as Easypaisa or JazzCash.
6. The system holds deposited payments in an **escrow state** until contractual conditions are fulfilled.
7. Freelancers can submit completed work and request payment release.
8. Clients can review submitted work and approve or request changes.
9. Upon client approval, the system automatically releases the escrow payment to the freelancer.
10. Users can raise disputes if disagreements occur regarding work or payment.
11. The admin can review disputes, evaluate evidence, and decide payment outcomes.
12. The system sends notifications to users for important actions such as contract creation, payment updates, and dispute decisions.
13. Users can view their contract history, payment records, and transaction status.
14. The admin can manage users, contracts, payments, and system settings through a secure dashboard.
15. The system maintains logs of all major activities for transparency and auditing.

Software / Tools needed:

- **Tools:**

Git and GitHub: GitHub provides a cloud based version control system that allows the entire team to collaborate on code, track changes, and merge updates safely without overwriting each other's work

Jira: Jira supports Scrum and Kanban methodologies, helping the development team plan sprints, assign tasks, and track progress efficiently throughout the FYP lifecycle.

- **Technology:**

React Js

Component based, fast to build, large ecosystem (UI libraries, form handling). Next.js adds server-side rendering and easy routing if you want SEO friendly pages (landing, docs).

Backend (ExpressJS + NodeJS): Lightweight, JavaScript end-to-end (same language as frontend), vast middleware ecosystem, good for building REST/GraphQL APIs.

PostgreSQL: PostgreSQL gives strong ACID guarantees, transactions, and relational integrity (useful for payments, contracts, audit logs)

Deployment Platform: ContracthubPk is deployed on free tier cloud platforms, with React applications hosted on Vercel, the backend on Render, and the database on MongoDB Atlas, making it cost effective and suitable for academic use.

Supervisor Name & Signature

